

MATH 372: COMBINATORICS

HOMEWORK 03

DUE FRIDAY, 9 OCTOBER, 10P.M., VIA GRADESCOPE

Remark. When you can start. Problems 1 and 2 use only what we have done through Monday, September 28. Problem 3 needs Wednesday the 30th and Friday, October 2; Problem 4(c) and (d) need Monday, October 5. As before, the set is released early so that the first half can be done first; do not wait.

Remark. Conventions. Graphs are finite, undirected, and loopless. For a subgroup $G \leq S_n$ acting on B_n , the quotient B_n/G is the poset of orbits with $o \leq o'$ iff some $x \in o$ lies below some $y \in o'$; for an orbit o , $v_o = \sum_{x \in o} x \in \mathbb{R}B_n$ is its orbit sum, and $\widehat{U}_i: \mathbb{R}(B_n/G)_i \rightarrow \mathbb{R}(B_n/G)_{i+1}$ is the operator $\widehat{U}_i(o) = \sum_{o'} c_{oo'} o'$ with $c_{oo'} = \#\{x \in o : x < y\}$ for any $y \in o'$, as in class. For a graph G with vertices $v_1, \dots, v_p$, $L = L(G)$ is the Laplacian (degrees on the diagonal, -1 for each edge, 0 otherwise), L_0 is L with one row and the same column deleted, and $\kappa(G)$ is the number of spanning trees.

Remark. On these solutions. Each problem is worth up to five points for mathematical content and up to two for writing. Write for another student in this course: say what you are going to show before you show it, justify each step, and use complete sentences. End your submission with a short *Methods and tools* note — three or four sentences on what software you used and for what, and how you checked anything a machine produced for you.

Problem 1 (necklaces). Let $\sigma = (1\ 2\ \dots\ n) \in S_n$ and $G = \langle \sigma \rangle \cong \mathbb{Z}/n$, acting on B_n . An orbit of G on k -subsets is a *necklace* with n beads, k of them black.

- (a) For $n = 6$, list the orbits rank by rank, drawing each necklace and recording the size of each orbit. (You should find rank sizes $1, 1, 3, 4, 3, 1, 1$.)
- (b) Still for $n = 6$, write the matrix of $\widehat{U}_2: \mathbb{R}(B_6/G)_2 \rightarrow \mathbb{R}(B_6/G)_3$ in the orbit bases, computing each coefficient $c_{oo'}$ from its definition, and verify that $\widehat{U}_2$ is one-to-one.
- (c) Now let $n = p$ be prime. Show that for $0 < k < p$ every orbit of G on k -subsets has exactly p elements. Deduce a formula for the number of necklaces with p beads, k of them black, and conclude that p divides $\binom{p}{k}$ for $0 < k < p$.
- (d) What does the theorem from class say about the largest antichain in B_6/G ? The theorem proves that a chain decomposition of B_6/G into symmetric chains exists; find one by hand, using your Hasse diagram from (a).

Problem 2 (the grid, again). Fix $1 \leq k < n$ and let $G = S_k \times S_{n-k} \leq S_n$ be the subgroup of permutations that map $[k] = \{1, \dots, k\}$ to itself. For $0 \leq a \leq k$ and $0 \leq b \leq n - k$ let

$$O_{a,b} = \{x \in B_n : |x \cap [k]| = a \text{ and } |x \setminus [k]| = b\}.$$

- (a) Show that the $O_{a,b}$ are exactly the orbits of G on B_n , and that $O_{a,b} \leq O_{a',b'}$ in B_n/G if and only if $a \leq a'$ and $b \leq b'$. Conclude that B_n/G is isomorphic to the grid poset of Problem 3 on Problem Set 2 (with the roles of a, b there played by $k, n - k$ here).
- (b) Compute the coefficients $c_{oo'}$ for $o = O_{a,b}$ and o' an orbit of the next rank, and write $\widehat{U}_i(O_{a,b})$ as a combination of orbits.

- (c) Take $k = 1$. Write $\widehat{U}_i$ as a 2×2 matrix for $1 \leq i < n/2$ and check directly that it is one-to-one. What happens for $i \geq n/2$, and why is that consistent with the theorem?
- (d) In one paragraph: the theorem from class reproves part (c) of Problem 3 on Problem Set 2. What did the problem-set proof provide that the theorem does not?

Problem 3 (Laplacians). (a) Write down L for the complete bipartite graph $K_{2,3}$, with the two vertices of the small side listed first. Compute $\det L_0$ twice, deleting a vertex from each side, and confirm the answer by counting the spanning trees of $K_{2,3}$ directly.

- (b) Let G be connected. Prove, *without* using the Matrix-Tree Theorem, that $\det L_0$ does not depend on which row and column are deleted. (Suggestion: let $\text{adj } L$ be the adjugate, so that $L \cdot \text{adj } L = \text{adj } L \cdot L = (\det L) I = 0$. Explain why the kernel of L is spanned by the all-ones vector $\mathbf{1}$, deduce that every column and every row of $\text{adj } L$ is a multiple of $\mathbf{1}$, and conclude that $\text{adj } L = cJ$ for a constant c , where J is the all-ones matrix. Then identify the diagonal entries of $\text{adj } L$.)
- (c) Let G be disconnected. Show that $\det L_0 = 0$ for every choice of deleted vertex, so the Matrix-Tree Theorem remains true, in the form “ $0 = 0$,” for disconnected graphs.
- (d) (*Deletion-contraction.*) For an edge e of G , let $G - e$ be G with e removed and G/e be G with e contracted to a single vertex (keeping any multiple edges that result). Prove that

$$\kappa(G) = \kappa(G - e) + \kappa(G/e),$$

and use this to compute $\kappa(F_n)$ for $n \leq 4$, where the *fan* F_n is a path on n vertices together with one more vertex joined to every vertex of the path. Guess the general pattern.

Problem 4 (spanning trees from eigenvalues). (a) Verify Cauchy–Binet by hand for $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ and $B = A^T$: compute $\det(AA^T)$ directly, and as the sum over the three 2-subsets of columns.

- (b) Show that the eigenvalues of $L(K_{m,n})$ are 0 and $m + n$, each once, m with multiplicity $n - 1$, and n with multiplicity $m - 1$. (Look for eigenvectors supported on one side of the bipartition whose entries sum to zero, and for eigenvectors constant on each side.) Deduce that $\kappa(K_{m,n}) = m^{n-1}n^{m-1}$, and check it against Problem 3(a).
- (c) The cube Q_n is n -regular and its adjacency eigenvalues are $n - 2i$ with multiplicity $\binom{n}{i}$, $0 \leq i \leq n$. Use the eigenvalue form of the Matrix-Tree Theorem to find $\kappa(Q_n)$ in closed form, and evaluate it for $n = 2, 3, 4$.
- (d) Let G_4 be the two-flip graph of Problem 1 on Problem Set 2, whose spectrum you computed there. It has two isomorphic connected components. Use the spectrum to find the number of spanning trees of one component, and say what graph that component is.

Bonus Problem (trees through an edge). Show that the number of spanning trees of K_n that contain a given edge is $2n^{n-3}$. (Count pairs (tree, edge of the tree) in two ways.) Deduce $\kappa(K_n - e)$, and check your answer for $n = 4$ with the Matrix-Tree Theorem.